Forgetting You Can Budget, Delete and Schedule: a (μ, γ, T) Retention Law and Exact Store-Level Deletion for a Physics-Structured Recurrent Memory

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

A memory that forgets is only useful if you can say *how fast*, *why*, and *what is actually gone*. In a physics-structured recurrent memory — a damped symplectic step of a learned Hamiltonian — forgetting is a budget in three numbers: the spectral mass μ of the stored direction, the damping γ and the temperature T. The headline is a damping optimum: retention half-life is non-monotone in friction, one V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$ ($\propto \mu^{-2}$ overdamped, a mass-independent floor underdamped), spanning ≈ 11 orders of magnitude in μ^2, holding on emergent (learned-MLP) as well as designed units, 3/3 seeds (dim 4, hidden 64, laptop-CPU), and reproduced by direct nonlinear rollout with the argmin agreeing to 0.35%. Hence at $T > 0$ friction preserves and temperature erases, and a localized friction hole is a memory vault: $107.77 \pm 4.78\times$ designed, $106.1 \pm 5.0\times$ law-referenced on emergent (3 seeds each), though the field-versus-scalar contrast naming its mechanism is designed-only. Removal is structural: under canonical placement — Blelloch & Golovin's (FOCS'07) — store-level deletion is exact, the post-deletion store byte-identical at every load measured to the store that never held the item, under three stated conditions (`budget >= n_cells`, `leak = 0`, priority/attribute eviction; recency eviction excluded), on a designed store of dim 3, capacity 8–64, no learning. We state what decay does *not* buy: the store stops answering before it stops leaking.

1 Introduction

Deployed agent memories forget, and forgetting is the failure mode benchmarks do not measure (Yang, 2026; Uddin et al., 2026). In a recurrent network forgetting is a side effect of gate saturation, a decaying eigenvalue or a mixing chain, rarely quantified in advance: one cannot read off a trained LSTM the half-life of a stored value, the knob that sets it, or what remains once the value is removed. We study a memory where all three are answerable. Its state (q, p) advances by a damped symplectic step of a learnable Hamiltonian $H = T(p) + V_\theta(q)$: the damping $p \mapsto (1-\gamma)p$ supplies forgetting, Langevin noise supplies temperature T, and items are optionally written as *atoms*: localized wells superposed into one energy function a dynamical relaxation reads. Our reference unit is the CLU (Causal Learning Unit), introduced as CHLU in Jawahar & Pierini (2026). Its core is exactly solvable: diagonalised in the mass-whitened Hessian, one damped step acts on each normal mode as a single 2×2 matrix fixed by $(\varepsilon\mu, \gamma)$, whose spectral radius is that mode's retention (a theory note develops this at length; Anonymous, 2026, on which nothing here depends).

Contributions. *(1) A (μ, γ, T) budget with a damping optimum* (§3.1–§3.2): retention is one non-monotone V-curve minimized at $\gamma_{\rm crit} = 2\varepsilon\mu$, so latch, overdamped register and underdamped working memory are three regimes of one curve, evaluated at two values of μ, not two laws — across ≈ 11 decades in μ^2, on emergent as well as designed units, on two independent instruments. At $T > 0$ friction preserves and temperature erases, and a friction hole is a vault whose two constituent laws hold on both arms. *(2) A composition, not a new deletion algorithm* (§3.3): the placement rule and its delete-time repair are Blelloch & Golovin's (FOCS'07); new here is only the composition — a lattice packing certificate, decaying content whose decay provably commutes with deletion, one canonical energy function read by a dynamical relaxation — plus the trilemma that prices lifetimes and an honest leakage statement. *(3)* A three-state memory lifecycle (PROTECTED $\rightleftarrows$ ACTIVE $\rightarrow$ TRASH, 7/7 legs, kill-conditions committed before the verbs, one leg unexercised), shipped as a mechanics demonstration on a toy synthetic build: no value or benchmark number is claimed, and none was run.

Nomenclature, because "mass" names two opposed quantities. The inertial mass M_i is the learned diagonal of the kinetic term (larger $\Rightarrow$ slower); the spectral mass is $\mu_k^2 = \lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$ (larger $\Rightarrow$ shorter memory). Retention statements use μ; a stored value lives on a near-flat coset direction; ε is the integrator step, $= 0.05$, never a tilt. Half-lives mean nothing without a read tolerance Δ and the thermal persistence length ℓ_θ, so we never quote an $n_{1/2}$ without its Δ and ℓ_θ/Δ, and never read a designed-versus-learned distinction off raw exponents. Results on designed testbeds are labelled verification of exactness; on learned or emergent potentials, evidence. Every $T > 0$ result requires the fluctuation–dissipation-consistent noise scale $\sigma_i^\star = \sqrt{M_i T \gamma(2-\gamma)}$ *and* a Newtonian kinetic mode. These laws govern the store; end-to-end performance additionally depends on the encoder, measured separately, φ-bytes ledgered.

2 Related work

Deployed long-term memories expire and delete by policy rather than by law: Zep invalidates contradicted edges with a timestamp rather than removing them (Rasmussen et al., 2025), Generative Agents weight recency by a 0.995 decay factor in a ranking heuristic (Park et al., 2023), Titans carries a learned forget gate (Behrouz et al., 2025), and Mem0 exposes the update phase as a DELETE tool call (Chhikara et al., 2025). We benchmark against none of them; the contrast is structural. Each mechanism is a score, a flag or a learned hyperparameter where damping here carries a retention law, a read tolerance and a temperature, and each deletion is a tombstone where ours is a byte-identity — which matters, since tombstoned vectors remain reconstructible from raw index files (Chakraborttii et al., 2026). Learned recurrent comparators (LSTM, Hochreiter & Schmidhuber, 1997; LEM, Rusch et al., 2022) lack a stateable forgetting law. In physics, dissipation turns type-A Nambu–Goldstone modes from propagating to diffusive (Minami & Hidaka, 2018) — exactly our coset — and Mo (2026) proves that an exactly G-equivariant flow has *at least* $\dim(G/\mathcal{H})$ zero Lyapunov exponents along the orbit: the kinematics of protection, to which our budget is the constitutive complement, since the gap is μ^2.

Order-independent placement is not ours either — it runs from Snyder (1977) through Naor & Teague (2001) to Blelloch & Golovin (2007), whose table we use, and Blelloch, Golovin & Vassilevska (2008) had already taken unique representation into computational geometry, so canonical placement in a geometric setting is not virgin ground (Appendix D). *Certified* removal is Guo et al.'s (2020) §2 Eq. (1), an ε condition with an unnumbered (ε, δ) relaxation immediately after, and and exact methods delete by isolation, with cost and capacity formalisms in place (Bourtoule et al., 2021; Ginart et al., 2019; Sekhari et al., 2021). We make none of those claims: ours is a store-level bit-exactness statement with the encoder excluded, coining no benchmark and no cost claim.

3 Results

3.1 The damping optimum: one V-curve across eleven decades in spectral mass

The step is conformally symplectic, $\det J = (1-\gamma)^d$, and $(1-\gamma\phi(q'))^d$ when position-gated; checkpoints (dim 4, hidden 64, $\varepsilon = 0.05$, 150 epochs) are *designed* ($SO(2)$-invariant V_θ, 5 seeds) or *emergent* (MLP V_θ on identical symmetric data, 3 seeds). Retention half-life is non-monotone in friction. On a trained designed-$SO(2)$ vacuum the massive radial mode's $n_{1/2}(\gamma)$ is a V minimized at $\gamma_{\rm crit} = 2\varepsilon\mu_{\rm rad}$, the overdamped branch following the μ^{-2} law up to $\varepsilon\mu \approx \gamma/2$ and then saturating at a mass-independent floor (log-slopes -1.006 and $+1.23$–$+1.27$; 5 seeds, verification, Appendix A). The flat coset (Hessian $\mu^2 \approx 10^{-15}$) is the *same curve at* $\mu \to 0$, where $\gamma_{\rm crit} \to 0$: latch, overdamped register and underdamped working memory are three regimes of one curve, evaluated at two values of μ — not two laws. It generalizes to emergent units (evidence): on three emergent MLP checkpoints trained on the same symmetric data the coset-tracked $T = 0$ half-life traces the same V-curve —

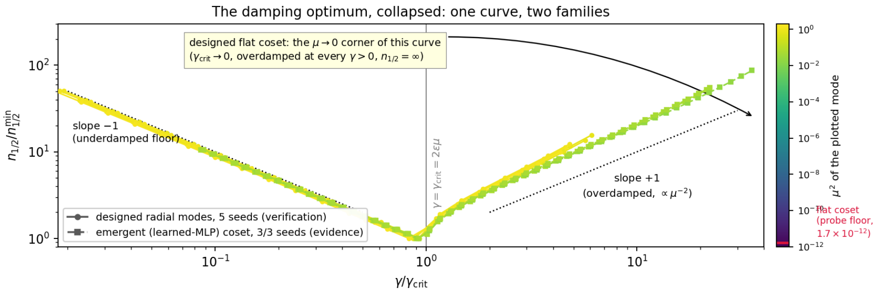

Figure 1: *The damping optimum, collapsed (headline).* Eight measured curves collapse onto one: five designed radial modes (circles, verification, $\mu^2_{\rm rad} = 0.670$–1.348) and three emergent learned-MLP coset modes (squares, evidence, $\mu^2_{\rm soft} = 2.0$–5.4×10^{-2}, 3/3 seeds), against the ∓ 1 asymptotes; the designed flat coset is the $\mu \to 0$ corner of the same curve. *(One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU;* `langevin_noise="fdt"`*, Newtonian kinetic mode.)*

argmin $0.902 \pm 0.003 \times \gamma_{\rm crit}$, log-slopes -1.0020 ± 0.0003 / $+1.116 \pm 0.011$, 3/3 seeds. Across both families the curve spans $\mu^2 \in [1.7 \times 10^{-12},\ 7 \times 10^{-2}]$, that low endpoint being the ring-profile probe's resolution on a checkpoint whose Hessian μ^2 is machine zero rather than a spectral mass — so eleven orders is one curve on one instrument, though the $\mu \to 0$ corner has its own model-side bound of $n_{1/2} \geq 9.5 \times 10^{15}$ steps by direct $T = 0$ rollout (Appendix B).

Two instruments, one law (evidence). Every point above is a one-step Jacobian, so we re-measured the curve by direct nonlinear rollout, reading $n_{1/2}$ from the decay envelope's rate (3 emergent seeds, $\delta = 0.05$–0.5 rad). The shape reproduces: argmin 0.9001 ± 0.0052 against the Jacobian's 0.9032 ± 0.0027 in units of $\gamma_{\rm crit}$ — a 0.35% difference — log-slopes -0.983 ± 0.013 / $+1.117 \pm 0.010$ on identical fit windows, and decay *rates* agreeing to 0.995 ± 0.003 in the asymptotic window. The instruments differ by a level, not a law: a first-crossing threshold reads short by a per-checkpoint constant ($1.33/1.80/1.82\times$) flat in γ, with coefficient of variation 2.2–3.7%. *What would have falsified the law:* a stored direction that mixes normal modes, or is anharmonic at the write amplitude, has no single-mode reduction, so the curves would neither collapse nor put their minimum at $2\varepsilon\mu$; neither happens at ten times the write amplitude, where the argmin moves by at most 1.4% (to 0.897–$0.917\,\gamma_{\rm crit}$) and the slopes by at most 1.9% (Appendix B).

3.2 Friction preserves, temperature erases — and a friction hole is a vault

At $T = 0$ a designed coset latches forever at any friction (drift $\leq 4.9 \times 10^{-12}$ rad over 200k steps, all $\gamma \in [0.002, 0.5]$, 5 seeds, verification): friction has no contractive action on a flat direction. At $T > 0$ the coset decays diffusively and friction slows it down — $D_\theta = \varepsilon T(2 - \gamma)/(2F^2\gamma)$ is verified to 1.0068 ± 0.0219 over 25 (γ, T) cells, so $n_{1/2} \propto \gamma^{+1}$ and the sign flip $\partial n_{1/2}/\partial\gamma > 0$ holds in 10/10 seed $\times$ temperature conditions, $\gamma : 0.05 \to 0.2$ *lengthening* the half-life $3.77 \pm 0.23\times$ at $\Delta = 0.5$ rad, $\ell_\theta/\Delta < 0.05$; only temperature has $\partial n_{1/2}/\partial T < 0$. *Fine print, beside the claim:* every $T > 0$ number requires `langevin_noise="fdt"` *and* a Newtonian kinetic mode; under the reference default these laws fail, and in the relativistic mode no noise scale targets Gibbs.

Hence a memory vault. The shipped position-gated friction field is *absorb-only*, so a hole is a brake *and* a refrigerator ($T_{\rm local} = 1.26 \times 10^{-4}$ against 10^{-3} outside, where a coupled bath predicts none), and the vault factor is $(\gamma_{\rm eff}/\gamma)^2$ — $107.77 \pm 4.78\times$ (D_θ-estimator, pre-registered, 3 designed seeds, $\Delta = 0.5$ rad) against $13.28 \pm 0.12\times$ for a scalar friction of equal $\gamma_{\rm eff}$, the ratio 8.11 ± 0.37 being the refrigerator. At $T = 0$ it erases nothing. Both laws transfer to an emergent register (evidence, 3 emergent seeds at $T = 4 \times 10^{-3}$ and 8×10^{-3}, both above $T^\star$): the refrigerator law holds at 0.9998 ± 0.0019 of prediction (the coupled bath rejected again at 0.2235) and the $\hat{D}_\theta \propto \gamma_{\rm eff}^{-2}$ law at 1.016–1.103, a law-referenced vault of $106.1 \pm 5.0\times$ beside the designed $107.77 \pm 4.78\times$. One result has no designed analogue — *the hole confines*: the fraction of states outside the register ($|\theta| > 1$ rad) falls from $5.5/43.0/2.4\%$ with no hole to 0.0000 inside it, 3/3 seeds, while a scalar friction of the same $\gamma_{\rm eff}$ still hops ($0.73/10.2/0.26\%$). *The contrast number stays designed-only:* on emergent the field/scalar ratio measures 23.39 ± 10.06 because the *control* arm delocalises there ($\hat{D}/D_{\rm flat} = 1.18$–$5.23$) while the field arm sits on the law, so 8.11 ± 0.37 is a designed quantity and the direction, not the number, transfers. We quote no emergent σ_θ ratio: above $T^\star$ that diagnostic is non-stationary here, its control returning 0.459 ± 0.118 where it must return 1.000 (Appendix C).

The designed-symmetry precondition, the boundary on everything above (evidence; 3 emergent seeds and a matched designed control). The exact $T = 0$ latch is a *designed* object: on an MLP potential trained on identical symmetric data the softest direction is a mid-spectrum massive mode ($1 - |\lambda_{\mathrm{coset}}| \approx 10^{-3}$ against the designed $\leq 1.1 \times 10^{-15}$), a written value relaxes completely, and capacity is $\approx$ 1–1.6 bits. An emergent unit of this class has no continuous coset register, and continuum flat directions must be designed in. What survives is exactly the headline — the V-curve on 3/3 seeds and the sign flip above a measured $T^{\star} \approx 3 \times 10^{-3}$. The same boundary bounds an extrapolation we do not make, and Appendix E records it.

3.3 Removal is structural: exact store-level deletion, priced lifetimes, honest leakage

(Designed atom store, dim 3, no learning anywhere ⇒ verification.) Stated once: external benchmarks won on their own headline metric = ZERO. A flat table deletes exactly by construction — the trivial substitute; at issue is a store whose items are *superposed into one energy function*, where placement would ordinarily depend on which others are present. Placement here is canonical, a function of the live records and the geometry alone, so removal reproduces the store holding the rest bit for bit. Measured: *"with the waitlist, store-level deletion is exact at every load measured,* $0.29\times$–$1.71\times$ *lattice capacity — AUC* 0.5000 ± 0.0000 *on all six statistics, byte-equal* 1.0000 *— under three stated conditions (`budget >= n_cells`, `leak = 0`, priority/attribute-based eviction; recency-based eviction remains intrinsically history-dependent and is excluded)."* The allocator it replaces leaks membership after eviction (AUC 0.99985 at full load), and the packing price of exactness is negative: canonical placement admits strictly more keys than refuse-and-relocate. This is a store-level guarantee only — the frozen encoder and any residue of past writes in a learned landscape are separate channels, measured separately, and we claim no certified (ε, δ) unlearning; without the waitlist, exactness fails at overflow (Appendix D).

Lifetimes are priced by a trilemma — exact value fidelity, amplitude-independent address hold, amplitude-independent read latency: pick two. The corner to lead with: *dropping amplitude-independent latency is the compute-adaptive-read dial — a faded memory costs more integration steps to read, a physical, measurable statement a timestamped row cannot make. Attribution, in the same breath as the theorems. Our placement rule is the Blelloch–Golovin (FOCS'07) stable-matching table, our Theorems 1–2 follow from theirs, and the fix-up cascade is theirs.* New here is only the composition — packing certificate, decaying content with a proven delete/decay commutation, one canonical energy function — whose price is negative where the general case can cost an exponential slowdown (Buchbinder & Petrank, 2003). That separation is between the *weak and strong* notions for comparison-based structures, whereas ours is against a history-*dependent* stochastic relocation rule, so the two statements are adjacent rather than identical.

What decay does not buy. The store stops answering before it stops leaking: at the last amplitude before self-eviction ($A = 0.051$) retention is 0.832 — placement-dependent, quoted for the controller-placed disk — while a query-level membership adversary sits at AUC 0.983 and a white-box one at 1.000; the curves never cross. Against an *exact* adversary decay reduces the effect size, not the AUC, so we claim no reduction in distinguishability per se; and the tighter laundering control fires — a boolean TTL flag *inside the same store* stays within 0.017 AUC of physical decay (0.983 versus 1.000), separating only against a resolution-limited adversary. The differentiator we lead with needs no adversary model: *decay contracts the addressing tolerance of an item,* R_{50} $1.146 \to 0.752$ *(*$1.52\times$*), where a TTL dict's lookup radius is a constant hard step at* 0.75–0.77*, independent of age.*

4 Limitations and future work

Limitations. (i) One architecture, one scale: dim 4, hidden 64, laptop-CPU, 5 designed / 3 emergent seeds; store at dim 3, no learning; lifecycle on a toy synthetic build. (ii) The designed-symmetry precondition (§3.2). (iii) Raw T/γ-exponents do not discriminate designed from emergent; the `fdt` + Newtonian scope is mandatory at every $T > 0$. (iv) The V-curve's shape is instrument-independent, its absolute step counts are not (a first-crossing threshold reads 1.33–$1.82\times$ short of the Jacobian, a constant per-checkpoint offset), and the emergent collapse spans only $2.7\times$ in μ^2 across three seeds, so the eleven decades are one decade of emergent variation attached to the designed $\mu \to 0$ corner. (v) Deletion is store-level only, under set-function eviction with recency eviction excluded, and no deletion-cost claim is made ($O(n)$ rebuild per operation; unrun). (vi) No task-level payoff is claimed. *Scale is a scope choice:* this short is sealed to laptop-scale evidence; no larger-scale result is quoted or should be inferred. *Named next:* a wider emergent collapse in μ^2; the vault's contrast arm re-measured where both arms stay bounded; amortized per-delete cost against the flat-datastore substitute; deletion at 10^3-item stores; and a localized *temperature* field.

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A The (μ, γ, T) budget: the diffusion law and the sign flip

Designed $SO(2)$ testbed $\Rightarrow$ verification. All $T > 0$ cells require `langevin_noise="fdt"` and a Newtonian kinetic mode; the reference default is `legacy`, under which none of these laws hold. Every $n_{1/2}$ is quoted with its read tolerance $\Delta = 0.5$ rad and its ℓ_θ/Δ.

Diffusion law, 25 cells ($5\gamma \times 5T$, seed 44): $\hat{D}_\theta/D_\theta^{\mathrm{pred}} = 1.0068 \pm 0.0219$ (min 0.9644, max 1.0484); $d\log D/d\log T = +0.998, +0.997, +0.992, +1.010, +1.006$; $d\log D/d\log\gamma = -1.014, -0.980, -0.999, -1.001, -0.982$ after dividing the discrete $(2-\gamma)$ factor. *Sign flip:* $d\log n_{1/2}/d\log\gamma = +0.9552 \pm 0.0422$ at $T = 10^{-3}$, 5 seeds, positive in 10/10 conditions; effect size $\gamma : 0.05 \to 0.2$ is $3.77 \pm 0.23\times$ (per-seed 3.55, 3.64, 3.61, 3.89, 4.15); $d\log n_{1/2}/d\log T = -0.956$ to -0.979 deep-diffusive. *Latch at $T = 0$, designed only:* $\big||\lambda_{\mathrm{flat}}| - 1\big| \leq 1.7 \times 10^{-14}$ over 22 γ values on 5 seeds, drift $\leq 4.9 \times 10^{-12}$ rad per 200k steps in 30/30 cells, against a $\gamma = 0$ control drifting 142.7 rad per 20k. *Designed V-curve:* argmin 0.076–0.096 against a predicted 0.082–0.116 (per-seed $\mu^2_{\mathrm{rad}} = 0.670, 0.771, 1.190, 1.093, 1.348$); asymptotic log-slopes -1.006 and $+1.23$–$+1.27$; mass-independent floor 27.03 steps at $\gamma = 0.05$.

Table 1: The $T > 0$ face of the budget, all four entries measured on the designed testbed.

stored direction	$\partial n_{1/2}/\partial\gamma$ (more friction)	$\partial n_{1/2}/\partial T$ (more heat)
coset / flat (designed)	> 0, friction *preserves* ($+0.955 \pm 0.042$; $3.77\times$ for $\gamma : 0.05 \to 0.2$)	< 0, temperature *erases* (-0.956 to -0.979)
massive (underdamped)	< 0, friction shortens (slope -1.006)	≈ 0; T sets only the fluctuation floor (max/min ≤ 1.013, 9/9)

B The emergent arm, and the V-curve on a second instrument

Emergent MLP checkpoints, 3 seeds, plus a matched designed control $\Rightarrow$ evidence. Pre-registered; predictions filed before any harness ran.

Four instruments, identical fit windows ($\gamma < \gamma_{\mathrm{crit}}/2.5$, $\gamma > 2.5\gamma_{\mathrm{crit}}$, keyed to the Jacobian's γ_{crit} so the windows are not a free parameter): *I-J*, the one-step Jacobian, $n_{1/2} = \ln 2/(-\ln|\lambda_{\mathrm{ret}}|)$; *I-R1*, rollout first crossing of $|\theta| \leq \delta/2$; *I-R2*, rollout last crossing; *I-R3* (primary), rollout envelope-rate fit over the late window, $R^2 \geq 0.9955$ (median 0.9996) on every cell.

Finite write amplitude. At $\delta = 0.05/0.2/0.5$ rad the I-R3 argmin is $0.8928/0.9007/0.8968$ (seed 42), $0.9047/0.9045/0.9174$ (43), $0.9028/0.9037/0.9070$ (44) — at most $+1.4\%$ over a $10\times$ range — and the slopes move by at most 1.9%; every $\delta = 0.5$ write relaxed all the way home ($|\theta_{\mathrm{final}}| \leq 6.5 \times 10^{-10}$), so no

Table 2: The shape claims on all four instruments ($\delta = 0.05$ rad, 48 γ values, 3 emergent seeds). The rollout rate instrument reproduces the Jacobian; both threshold instruments fail below γ_{crit}, so no rollout $n_{1/2}$ is ever quoted from a threshold there.

	$\gamma_{\mathtt{crit}}$	argmin/$\gamma_{\mathtt{crit}}$ (I-J / I-R1 / I-R2 / I-R3)	slope below
seed 42	0.02334	0.8994 / 0.0857 / 0.2068 / 0.8928	-1.0023 / $+0.0725$ / -1.0556 / -0.9651
seed 43	0.01424	0.9046 / 0.1404 / 0.1404 / 0.9047	-1.0016 / $+0.0688$ / $+0.0688$ / -0.9977
seed 44	0.02265	0.9055 / 0.0883 / 0.3031 / 0.9028	-1.0022 / $+0.0455$ / -0.8854 / -0.9854
mean $\pm$ sd		0.9032 ± 0.0027 / 0.1048 ± 0.0252 / 0.2168 ± 0.0668 / 0.9001 ± 0.0052	-1.0020 ± 0.0003 / $+0.0623 \pm 0.0120$ / -0.6240 ± 0.4948 / -0.9

Table 3: The instrument gap is a level, not a rate: on the overdamped branch ($\gamma \geq 2\gamma_{\mathrm{crit}}$) the decay *rates* agree while the *threshold* instrument is offset by a per-checkpoint constant flat in γ. Seed 44's full-window value (0.9216 ± 0.0027) sits 0.0004 below its pre-registered band because that window still contains a second, faster mode (early-window ratio 0.539); the window-split diagnostic converges monotonically to 1 into the asymptote. Cause of the threshold offset, measured: the ring write deposits only a fraction of its amplitude in the slow branch (envelope-fit intercept over δ = $0.638/0.644/0.327$).

seed	$\Gamma_{\mathrm{jac}}/\Gamma_{\mathrm{R3}}$ (asymptotic window)	$n_{\mathrm{jac}}/n_{\mathrm{R1}}$ (threshold)	CV of $n_{\mathrm{jac}}/n_{\mathrm{R1}}$ over the grid
42	0.9935	1.3307	
43	1.0000	1.7981	2.2–3.7% over 21–25 γ-points
44	0.9948	1.8204	
mean	0.995 ± 0.003	per seed, at $\delta = 0.05$ rad	$\lvert d\log(\mathrm{ratio})/d\log\gamma\rvert \leq 0.0034$ (threshold 0.10)

seed's write cleared its washboard barrier. The pre-registered early-window clause *fails*, $\Gamma_{\mathrm{jac}}/\Gamma^{\mathrm{early}}_{\mathrm{roll}} = 0.52$–$0.85$ against a band of $[0.65, 1.35]$; the amplitude sweep identifies that as mode mixing rather than anharmonicity, seed 44's early ratio being $0.539/0.527/0.522$ across the same $10\times$ range.

What the left branch does not contain. For $\gamma < \gamma_{\mathrm{crit}}$ the coset eigenpair is complex with $|\lambda|^2 = 1 - \gamma$, so $n_{1/2} = \ln 2/(-\frac{1}{2}\ln(1-\gamma))$ *independently of* μ^2 *and of the checkpoint*: at $\gamma = 0.002$, emergent seeds 42 ($\mu^2 = 5.449 \times 10^{-2}$) and 43 ($\mu^2 = 2.029 \times 10^{-2}$) both print $\lambda_{\mathrm{ret}} = 0.998999499$ and $n_{1/2} = 692.5$, against $2\ln 2/0.002 = 693.1$. All model-dependent content of the V-curve is the argmin location $\gamma_{\mathrm{crit}} = 2\varepsilon\mu$ and the $+1$ branch; what the rollout adds is that the *nonlinear* model obeys that identity, which was not guaranteed.

Designed control, and the model-side bound. Over 5 designed seeds $\times$ 48 γ $\times$ 3 write amplitudes $\times$ 32 000 steps of direct $T = 0$ rollout, $\max_n |\theta(n) - \delta| = 9.400 \times 10^{-13}$ rad, so $\Gamma \leq 7.309 \times 10^{-17}$ per step and $n_{1/2} \geq 9.483 \times 10^{15}$ steps against a pre-registered bound of 10^{-6} rad; the matched Hessian μ^2_{soft} there is 1.60×10^{-15}, 1.52×10^{-15}, -2.92×10^{-16}, 2.94×10^{-16}, 1.21×10^{-16}. *Collapse:* in $x = \gamma/\gamma_{\mathrm{crit}}$, $y = n_{1/2}\gamma_{\mathrm{crit}}$ the three emergent curves agree to a median 1.15% on I-J and 5.67% on I-R3; $y_{\min} = 1.579/1.527/1.499$ and $1.631/1.535/1.662$ sit 22–24% below the continuum $2\sqrt{2}\ln 2 = 1.961$ and $x_{\min} \approx 0.90$ sits 27% above the continuum 0.707, both known discrete-map corrections (matched designed control $x_{\min} = 0.83$–0.93). *Scope:* μ^2_{soft} spans only 2.029×10^{-2}–5.449×10^{-2} across the three seeds, a factor 2.7, so the eleven-decade span is one decade of emergent variation attached to the designed $\mu \to 0$ corner and the rollout does not widen it.

The emergent arm's own negative. $1 - |\lambda_{\mathrm{coset}}| = 1.06 \times 10^{-3}$–$2.96 \times 10^{-3}$ at $\gamma \approx 0.048$ against a designed $\leq 1.1 \times 10^{-15}$, with a whole-grid emergent minimum of 7.6×10^{-5}–2.0×10^{-4} against a designed grid minimum of exactly 0. Written $\delta \in \{0.1, 0.3, 0.5\}$ rad relaxes completely ($|\mathrm{retained}| \leq 2.1 \times 10^{-3}$) where the designed checkpoint retains δ exactly; capacity is $\approx$ 1–1.6 bits, not a continuum. Bias-corrected $n^{\mathrm{obs}}_{1/2}/n^{\mathrm{Goldstone}}_{1/2}$ at $\gamma = 0.2$, $\ell_\theta/\Delta < 0.06$, runs $2.75, 3.40$ at $T = 2 \times 10^{-3}$ to 0.94–1.25 at $T \geq 4 \times 10^{-3}$, giving $T^\star \approx 3 \times 10^{-3}$ (predicted 2.72–3.66×10^{-3}) against a washboard barrier of 2.29–3.57×10^{-2}. *Reporting rule:* raw exponents do not discriminate — a matched designed control gives $d\log n_{1/2}/d\log T = -0.53, -0.60, -1.04$ and $d\log n_{1/2}/d\log\gamma = +0.78, +0.63, +0.55$, indistinguishable from emergent, because both families carry the ℓ_θ/Δ boundary layer (up to 2.03). Only $|\lambda_{\mathrm{coset}}|$ and bias-corrected deep-diffusive cells discriminate.

C The friction-hole vault, on both arms

Designed $SO(2)$ plus an analytic field, 3 seeds $\Rightarrow$ verification; emergent MLP, 3 seeds $\Rightarrow$ evidence. Pre-registered on both arms. All cells `langevin_noise="fdt"`, Newtonian kinetic mode, $\Delta = 0.5$ rad.

Mechanism. The shipped field is absorb-only, $p \leftarrow (1-\gamma_\phi)(1-\gamma)p + \sigma(\gamma)\xi$ with $\sigma^2 = MT\gamma(2-\gamma)$ set by the *scalar* γ, so with $\gamma_{\mathrm{eff}} = 1 - (1-\gamma)(1-\gamma_\phi)$, $T_{\mathrm{local}} = T\gamma(2-\gamma)/[\gamma_{\mathrm{eff}}(2-\gamma_{\mathrm{eff}})]$ and $D_\theta = \varepsilon T\gamma(2-\gamma)/(2F^2\gamma^2_{\mathrm{eff}})$: a vault of $(\gamma_{\mathrm{eff}}/\gamma)^2$. A coupled bath would give $D_\theta \propto \gamma^{-1}_{\mathrm{eff}}$ and a vault of $[\gamma_{\mathrm{eff}}(2-\gamma)]/[\gamma(2-\gamma_{\mathrm{eff}})] = 13.88\times$ at $\gamma = 0.05$, $\gamma_\phi = 0.5$. The hypotheses differ by $7.942\times$; the coupled-bath value is a refuted prediction and never the vault.

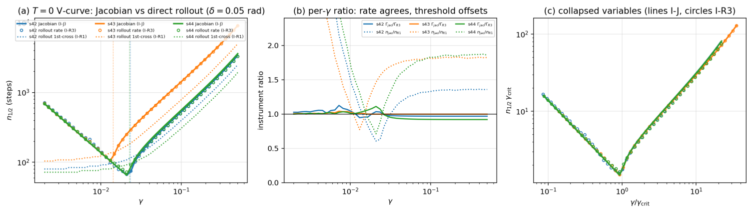

Figure 2: The V-curve measured a second way. (a) $T = 0$ $n_{1/2}(\gamma)$ per emergent seed on the one-step Jacobian (I-J, lines), the rollout envelope-rate instrument (I-R3, circles) and the rollout first-crossing threshold (I-R1, dotted), at $\delta = 0.05$ rad. (b) The per-γ instrument ratio: the rate ratio sits on 1, the threshold ratio is offset and flat in γ above γ_{crit}. (c) The same data in collapsed variables. *(Emergent MLP, 3 seeds, dim 4, hidden 64, $\varepsilon = 0.05$, float64, $T = 0$, pre-registered ⇒ evidence.)*

Designed arm. $\mathrm{Var}(p_i)/(M_i T) = 0.9986$ ($\gamma_\phi = 0$), 0.36171 ± 0.00129 (0.1), 0.23044 (0.2), 0.17513 (0.3), 0.12600 ± 0.00031 (0.5; absorb prediction 0.12591), so $T_{\text{local}} = 1.26 \times 10^{-4}$, a $7.94\times$ refrigerator. $\mathrm{MSD}/\mathrm{pred}_{\text{absorb}} = 1.0011 \pm 0.0215$ over 15 cells; the D_θ-based vault is $8.42\times$, $22.39\times$, $44.31\times$, $107.77 \pm 4.78\times$ at $\gamma_\phi = 0.1/0.2/0.3/0.5$ (absorb prediction 110.25). The direct first-passage vault reads $86.97 \pm 2.94\times$ and is boundary-layer biased on the outside arm ($\ell_\theta/\Delta = 0.079$), so $107.77\times$ is the quoted number and travels with its estimator's name. Scalar control at the same γ_{eff}: $13.28 \pm 0.12\times$ with $\mathrm{Var}(p) = M_i T$ intact, field/scalar 8.11 ± 0.37 against a predicted 7.942. At $T = 0$ the hole erases nothing (drift $\leq 1.75 \times 10^{-12}$ rad per 200k steps inside and outside, 12/12 cells; $\big||\lambda|_{\max} - 1\big| \leq 2.0 \times 10^{-15}$ over 27 cells including the hole edge) and attenuates the *write* instead: $\theta_{\text{settle}}(\text{in})/\theta_{\text{settle}}(\text{out}) = 0.09576$ against the transport law's 0.095238.

Emergent arm. The coset there is a bounded mode ($\mu^2_{\text{adiab}} = 2.77 \times 10^{-2}$–$8.93 \times 10^{-2}$), so the designed block estimator is biased low by up to 55% and is replaced by a window-free Ornstein–Uhlenbeck fit $\mathrm{MSD}(n) = A(1 - e^{-n/\tau})$, $\hat{D} = A/(2\varepsilon\tau)$, cross-validated on designed before use ($112.58 \pm 1.09\times$ against the banked $107.77 \pm 4.78\times$; $14.16 \pm 1.38\times$ against $13.28 \pm 0.12\times$; 8.03 ± 0.80 against 8.11 ± 0.37).

Table 4: The refrigerator law transfers exactly to an emergent register: $\mathrm{Var}(p_i)/(M_i T)$ against the absorb-only prediction, 2048 walkers × 40 samples, 3 emergent seeds; over all 24 field cells observed/absorb $= 0.9998 \pm 0.0019$, and at $\gamma_\phi = 0.5$ $T_{\text{local}}/T = 0.12570$, a $7.955\times$ refrigerator against 7.942 predicted. The coupled bath predicts 1.0 everywhere and is measured at 0.2235.

T	γ_ϕ	γ_{eff}	absorb pred.	coupled pred.	measured	obs/absorb
4×10^{-3}	0.0	0.050	1.00000	1.0	0.99815 ± 0.00257	0.9981
4×10^{-3}	0.1	0.145	0.36249	1.0	0.36326 ± 0.00067	1.0021
4×10^{-3}	0.2	0.240	0.23082	1.0	0.23062 ± 0.00080	0.9991
4×10^{-3}	0.3	0.335	0.17480	1.0	0.17473 ± 0.00046	0.9996
4×10^{-3}	0.5	0.525	0.12591	1.0	0.12592 ± 0.00055	1.0001
8×10^{-3}	0.5	0.525	0.12591	1.0	0.12548 ± 0.00032	0.9966

Table 5: The hole confines an emergent register, and the laundering control shows friction alone does not. $T = 4 \times 10^{-3}$, 1024 walkers, burn-in 20 000–80 000 steps, spread by interquartile range; "hop" means $|\theta| > 1$ rad. The pre-registered stationary-spread comparison is *void* here and no in/out spread ratio may be quoted from these checkpoints: the control that must return exactly 1.000 returns 0.4586 ± 0.1181.

seed	arm	γ_{eff}	σ_θ (IQR)	hop fraction	Boltzmann at T_{local}	obs/pred
42	no hole, $\gamma = 0.05$	0.050	0.40795	5.50%	0.24073	1.6946
42	scalar 0.525 (control)	0.525	0.20034	0.73%	0.24073	0.8322
42	γ_ϕ hole	0.525	0.06442	0.0000	0.08542	0.7542
43	no hole	0.050	1.17601	42.97%	0.46059	2.5533
43	scalar 0.525	0.525	0.35335	10.20%	0.46059	0.7672
43	γ_ϕ hole	0.525	0.08628	0.0000	0.16343	0.5279
44	no hole	0.050	0.36922	2.36%	0.24034	1.5363
44	scalar 0.525	0.525	0.21574	0.26%	0.24034	0.8977
44	γ_ϕ hole	0.525	0.07279	0.0000	0.08528	0.8535

The diffusion law on the emergent arm. On bounded cells, $\hat{D}_{\text{ou}}/D_{\text{absorb}} = 1.0480$ ($\gamma_\phi = 0$, 1 cell), 1.1026 ± 0.0799 (0.1, 4), 1.0158 ± 0.0519 (0.2, 5), 1.0552 ± 0.0496 (0.3, 5), 1.0418 ± 0.0519 (0.5, 6); against the coupled bath the same cells give 0.1312 ± 0.0065 at $\gamma_\phi = 0.5$. Law-referenced vault $106.1 \pm 5.0\times$ (6 cells; prediction $110.25\times$; designed $107.77 \pm 4.78\times$). A measured/measured ratio over the same cells reads

$297.8 \pm 196.8\times$ (median $234.4\times$); it is recorded and is never the vault number, because 5 of the 6 $\gamma_\phi = 0$ cells fail the bounded test ($\sigma_\theta^{\mathrm{ou}} = 0.42$–$885$ rad against a Boltzmann 0.24–0.65), while every $\gamma_\phi \geq 0.2$ cell passes.

The contrast number is designed-only, and the failure mode is the control. A scalar γ does not cool — $\mathrm{Var}(p)/(MT) = 0.99288 \pm 0.02354$ at $\gamma = 0.525$ with no field, where both hypotheses predict 1.0, against 0.126 for the field at the same γ_{eff} — so the mechanism statement (a friction field is not locally stronger friction; it is locally stronger friction *plus* a local refrigerator) stands on emergent. The contrast measures 23.39 ± 10.06 (per-cell $9.14/25.75/26.96/21.70/41.31/15.48$) against a pre-registered band $[6.5, 9.5]$ and a falsifier at > 14: it fired. The cause is the control arm, whose own $\hat{D}_{\mathrm{ou}}/D_{\mathrm{flat}}$ is $1.1768/3.7319/3.3782/2.7999/5.2284/2.0506$ — at $\gamma = 0.525$ with no cooling the emergent coset is still anharmonically delocalised — while the field arm sits on the absorb law at 1.02–1.15. The *direction* transfers; the *number* 8.11 ± 0.37 is a designed quantity.

First passage, and the confound it exposed. An emergent potential has no rotational symmetry, so a $\theta = \pi$ "outside" arm is not a vacuum: $|\nabla V| = 0.0420/0.1049/0.0414$ there against 0.0000 at $\theta = 0$, $V(\pi) - V(0) = +0.03163/+0.06003/+0.04315$, and μ^2_{adiab} differing by up to $2.15\times$ between the sites. Any emergent first-passage number quoted against a $\theta = \pi$ baseline is confounded and a same-site baseline is required. Same-site vaults ($|\Delta\theta| \geq 0.5$ rad, $T = 4 \times 10^{-3}$, $\gamma = 0.05$, 256 walkers, hole radius 1.0): $> 1379\times$ (seed 42), $35.5\times$ (43), $> 1290\times$ (44), the first and third being censoring-limited *lower bounds* (86.7% and 93.4% censored at a 10^6-step cap; closing them needs $\approx 10^7$ steps per walker). The pre-registered $> 300\times$ holds 2/3 and the falsifier fires on seed 43, already flagged as geometrically atypical — softest μ^2, shallowest barrier (6.8×10^{-3}), coset at $\sigma_\theta/\Delta = 0.92$ outside and 0.33 inside, never leaving the diffusive regime. The confound's signature is seed 44, whose $\theta = \pi$ first passage (15585) is $20\times$ its own $\theta = 0$ value (775) and whose scalar control reads $0.27\times$: more friction apparently *speeding* escape, impossible for a stationary register.

Scope carried by the whole emergent arm. Three model seeds; a single PRNG stream per cell, so every spread is across model seeds, not noise realizations; both temperatures above $T^\star$ while the designed study's trustworthy-cell restriction is $T \leq 3 \times 10^{-3}$, so every measurement is squeezed between $T^\star$ and delocalisation and only cells passing the bounded test support a law claim. The hole is a ball of radius 1.0 in the full 4-D space, so excursions of order 1.0 ($\approx 4.5\sigma_r$) leave it — a plausible contributor to seed 43's low vault, not separately measured. No temperature field was built.

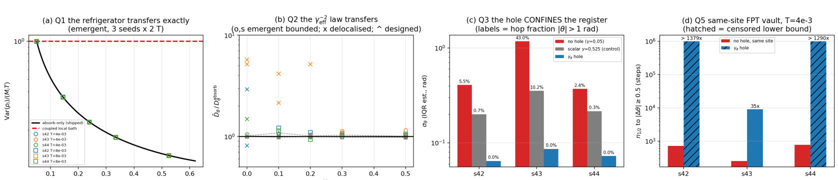

Figure 3: The vault on an emergent register; panels are labelled by the pre-registered prediction each tests. (a) The refrigerator law, $\mathrm{Var}(p_i)/(M_i T)$ against γ_{eff} for 3 seeds $\times$ 2 temperatures, on the absorb-only curve with the coupled-bath prediction (dashed) rejected. (b) The $\gamma_{\mathrm{eff}}^{-2}$ diffusion law: bounded cells as circles and squares, delocalised cells as crosses, the designed cross-check overlaid. (c) Confinement: stationary spread and hop fraction for no hole, the scalar control and the hole. (d) Same-site first passage; hatched bars are censored lower bounds. *(Emergent MLP, 3 seeds, $T \in \{4, 8\} \times 10^{-3}$, both above $T^\star$, $\Delta = 0.5$ rad $\Rightarrow$ evidence.)*

D Exact store-level deletion: prior art and the measured tables

Designed atom store, dim 3, capacity 8–64, no learning anywhere $\Rightarrow$ verification of the mechanism's exactness. Every deletion statement carries its scope: store level only, under the three stated conditions, with recency-based eviction permanently excluded.

Prior art, in full. Uniquely represented (canonical) data structures date to Snyder (1977) and Andersson & Ottmann (1995); *obliviousness*, a distributional condition on the memory representation which explicitly does not require canonical representation, is Micciancio's (1997); the weak and strong history-independence notions used here are Naor & Teague's (2001); the open-addressed realisation is Blelloch & Golovin's (2007), whose table is a stable matching between keys and slots under a global key priority, and it is that priority-greedy rule and that delete-time fix-up cascade we use; Blelloch, Golovin & Vassilevska (2008) had already taken unique representation into computational geometry. We claim no priority over order-independent placement and no novelty for the displacement rule or its delete-time repair. New is the composition: slots that are a lattice packing, so the canonical representation carries a minimum-separation certificate; content that is a continuously decaying amplitude whose decay commutes with deletion; a canonical object that is an energy function a dynamical read relaxes into; and a price that is negative where strong history independence is known to cost as much as an exponential slowdown relative to the weak notion for comparison-based structures (Buchbinder & Petrank, 2003). “Fix-up cascade” is our name for the `NEXT`-chain delete-time repair of their Figure 1; the algorithm is theirs either way.

Exactness, measured at zero tolerance. 24 exhaustive orders at $n = 4$, 200 random orders at $n \in \{8, 16, 40, 64\}$, 100 write/delete interleavings and mid-decay deletes: all bit-identical, minimum spacing exactly 1.540000; on the shipped store the centres, payloads, amplitudes and active mask are byte-equal and $V(q)$ is byte-equal on 64 random queries. *The packing price is negative:* at $K = 64$ the canonical rule admits $61/64$ offered keys deterministically ($\sigma = 0$), per-offered 0.9531, rising to $64/64$ when the sizing is inflated by $\times 1.05$, where the stochastic refuse-and-relocate allocator admits $43/64$ on the same seeds (per-offered 0.6719); delete-time churn is 2.836 moves per delete at full load. *At overflow, without the waitlist, exactness fails:* with 7 cells and 8 offers, $\mathrm{AUC}(n_{\mathrm{live}}) = 1.00000 \pm 0.0000$, white-box address-depth $\mathrm{AUC} = 0.91409 \pm 0.0813$, byte-equal fraction 0.0000. A pre-registration that failed informatively: $\mathrm{AUC}(z_{\mathrm{hole}}) \geq 0.85$ was registered there and measured 0.576, because the fix-up cascade *fills* the freed cell with a re-placed survivor.

Table 6: The laundering control that fires. Designed store, shipped two-phase read, no learning anywhere; 8 targets $\times$ 3 seeds, 128 paired worlds per target, 18 amplitude levels. Against an *exact* adversary a boolean TTL flag inside the same store is as detectable as an $e^{-\lambda t}$ amplitude, so there is no measurable differentiator at the AUC level; the control fails to fire only against a resolution-limited adversary, and that resolution is our own modelling choice — which is why the main text leads with the retrieval geometry instead.

adversary	physical decay	TTL flag in the same store	separation
white-box, exact	1.000	1.000	0.000
query-level membership, exact	0.983	1.000	0.017
query-level membership, $\sigma_{\mathrm{obs}} = 0.1$	0.559	0.996	0.437

What decays and what does not. Against an exact adversary the white-box per-example AUC is 1.000 at all 18 amplitude levels down to the floor; what decays is the effect size, the depth gap being $0.3935 \cdot A$ to four decimals at every level (the pre-registered closed form $1 - e^{-1/2}$) and d' falling $1\,560 \to 79.6$, exactly linear in A — the pre-registered falsifier fired on the AUC metric, as registered. Against a resolution-limited adversary the grading is real and the pre-registered law survives, $A_{75} = 2.57\sigma_{\mathrm{obs}}$ measuring $0.0780/0.2627/0.7537$ at $\sigma_{\mathrm{obs}} = 0.03/0.1/0.3$ against predicted $0.0771/0.2570/0.7710$; but σ_{obs} is our modelling choice, so its defensible use is mechanistic, never protective. *State the $|c|$ distribution with any retention number:* every retention magnitude here is for the controller-placed disk (proposal radius 2.2869, packing bound 8.00); on a max-radius ring the same store gives retention 0.500 at $A = 0.06$ where this placement gives 0.886. *Retrieval geometry:* R_{50} contracts $1.146/1.083/0.979/0.874/0.752$ as $A : 1 \to 0.5 \to 0.2 \to 0.1 \to 0.06$, matching the pre-registered saddle prediction at all five points (± 0.20), where a TTL vector store’s lookup radius is a constant hard step at 0.75–0.77.

The trilemma, and what it costs. The shipped store drops amplitude-independent address hold, which is why effective lifetime correlates $r = -0.85$ with a_i^2; the recommended gated-stiffness channel drops amplitude-independent latency and is the compute-adaptive-read dial — and it is *not* the shipped arm. Both previously proposed repairs are refuted: multiplying payloads by amplitudes kills the value criterion at $A = 1 - \mathrm{tol}/|a_i|$ (measured death amplitudes $0.90/0.80/0.70/0.30$ against the formula’s $0.900/0.860/0.767/0.300$) and inverts the payload dependence; launching the payload coordinate at the address-mapped point breaks the anti-decoration guard, after which a trivial substitute passes 0.672 of queries with no dynamics at all. Behind its flag the gated channel contracts R_{50} $1.135 \to 0.771$, retention std $0.274 \to 0.016$, worst codeword $0.398 \to 0.972$; the registered payload-independence test *fails as a correlation* ($|r|$ only $0.846 \to 0.667$) and passes as an effect size; and at $\gamma_{\mathrm{read}} = 0$ it is oscillatory, period $2\pi/\sqrt{\kappa(g_0 + A)}$ (log-log slope -0.5264), so no settling law may be quoted without a nonzero read friction. *No deletion-cost statement is made:* the shipped canonical synchronisation is an $O(n)$ rebuild per operation and the amortized-cost experiment has not been run.

E Prominent negatives

Every negative below is on the record with its measurement; none is dropped, and each anchors a future-work item.

claim tested	result
A continuous coset register emerges from training	No: an MLP potential on symmetric data yields a mid-spectrum massive mode (1.7–$4.9\times$ softer than the stiffest), capacity ≈ 1–1.6 bits, complete relaxation of any written value. The $T = 0$ latch is designed-only.
Raw T/γ-exponents discriminate designed from emergent	No: a matched designed control reproduces the same shallow slopes; only ℓ_θ/Δ-controlled cells and $\lvert\lambda_{\mathrm{coset}}\rvert$ discriminate. Never an $n_{1/2}$ without Δ and ℓ_θ/Δ.
The friction hole thermalizes locally (coupled bath)	Rejected by a pre-registered measurement, by a factor 8.11: the field is absorb-only and the vault is $107.77 \pm 4.78\times$, against $13.28 \pm 0.12\times$ for the scalar control.
A friction field can *forget* a coset	No: at $T > 0$ it *lengthens* coset memory. Erasing latched content needs a temperature or tilt field — which explains the observed failure of a learned friction field to recover fit on a stationary vacuum.
Our own sign-flip prediction, in both observables	Falsified in one: a register written at the washboard minimum cannot flip back, and both signs coexist on one checkpoint for different observables. A sign-flip figure that does not name the initial condition is meaningless.
Amplitude decay reduces an exact adversary's distinguishability	No, at any point in an item's life: white-box AUC 1.000 at all 18 amplitude levels; what decays is the effect size. The graded curve exists only relative to a stated adversary resolution, our own modelling choice.
Canonical placement is exact at overflow without the waitlist	No: $\mathrm{AUC}(n_{\mathrm{live}}) = 1.000$, byte-equal fraction 0.0000. Every deletion sentence carries its scope clause; recency eviction is permanently excluded.
The placement algorithm is ours	No: Blelloch & Golovin (2007) own it outright, and unique representation in a geometric setting is also taken (2008). Our claim is the composition only.
A per-deletion cost claim can be made	No: the shipped synchronisation is an $O(n)$ rebuild per operation, so the amortized-cost formalism cannot be instantiated on our own code until that experiment runs.
The payload-lifetime coupling has a repair	Both proposed repairs are refuted, and the registered payload-independence statistic fails as a correlation while passing as an effect size. At zero read friction the gated channel is oscillatory, not relaxational.

claim tested	result
A pseudo-Goldstone tilt is a lifetime dial on a learned store	Refuted *in sign*: a tilt monotonically *reduces* the smallest eigenvalue on two independent implementations and every family, because a written site is not a common vacuum of its own atoms. The relation survives only in the single-atom geometry it was specified in.
A tilt sets the soft scale on a learned store	No: the confinement floors the soft eigenvalue at 2α, capping lifetime at $\tau_{\mathrm{max}} = \Gamma/2\alpha$. The registered damped-mode floor is confirmed, the $1/$tilt branch unreachable, and lowering the confinement breaks the write.
The lifecycle's protected-fraction leg is exercised	No: 0 refusals on the stream at the measured operating point. Reported as *unexercised* — not working and not broken.
A task-level payoff is claimed	No: forgetting here is shown predictable, controllable, removable and schedulable, not shown to improve a downstream metric. External benchmarks won on their own headline metric = ZERO.
The field/scalar vault contrast transfers to an emergent register	No: 23.39 ± 10.06 against 8.03 ± 0.80 from the identical estimator on designed; the pre-registered falsifier fired. The cause is the laundering control itself, off its own law by 1.18–$5.23\times$. The number is designed-only; its direction transfers.
The emergent stationary-spread diagnostic is measurable above $T^\star$	No: its own control, which must return 1.000, returns 0.4586 ± 0.1181 — the no-hole arm is a hopping rotor, so no in/out spread ratio may be quoted. The confinement statement replaces it.
A $\theta = \pi$ baseline is valid for emergent first passage	No: $\lvert\nabla V\rvert = 0.042$–$0.105$ there against 0.0000 at $\theta = 0$, and one seed's scalar control reads $0.27\times$, i.e. more friction apparently speeding escape. Same-site baselines are required; the designed arm is protected by symmetry but was never checked, and that check is owed.
Threshold instruments are usable below γ_{crit}	No: a first-crossing $n_{1/2}$ puts its argmin at the grid edge with a below-slope of $+0.0623 \pm 0.0120$ where the law says -1, and last-crossing carries an additive half-period bias. Only the envelope-rate instrument is quotable there.
Our microscopic explanation of the instrument offset	Wrong, though the offset itself is real, constant and quotable: the pre-registered single-slow-mode projection fails (amplitude fraction $0.638/0.644/0.327$ does not order like the angular overlaps) and the early-window ratio 0.52–0.85 is amplitude-independent — multi-mode transient, not anharmonicity.
The emergent first-passage vault is measured	Only bounded: $> 1379\times$ and $> 1290\times$ at 86.7%/93.4% censoring, and $35.5\times$ on the geometrically atypical seed. Three seeds cannot say which is typical.

Anonymization note. Any supplementary or linked material, including code, is anonymized; only an anonymized snapshot may be linked.